

# Software Engineering and Project Management Lab Experiment No: - 10

## Aim: To Study and Implement Dockerfile instructions

**Aim:** To learn Dockerfile instructions, build an image for a sample web application using DOCKERFILE.

### **Theory:**

A container is a standard unit of software that packages up code and all its dependencies so the application runs quickly and reliably across different computing environments. Docker provides a platform for developing, shipping, and running containers easily.

#### **Key Features of Containers:**

- ❖ Isolation: Each container runs in its own isolated environment.
- ❖ Portability: Containers can run consistently across any system.
- ❖ Lightweight: Share OS kernel, making them more efficient than virtual machines.
- ❖ Scalable: Easily deploy and replicate containers across clusters.
- ❖ Fast Deployment: Start in seconds, ideal for CI/CD and agile development.
- ❖ Reproducible: Same environment from development to production.

#### **Docker as a Container Platform**

Docker enables you to create and manage containers using simple commands. It allows you to containerize applications and services with ease and deploy them in any environment.

#### **Key Docker Container Components:**

| Component        | Purpose                                       |
|------------------|-----------------------------------------------|
| Dockerfile       | Blueprint to build a Docker image             |
| Docker Image     | Snapshot of the app and its dependencies      |
| Docker Container | Runnable instance of a Docker image           |
| Docker CLI       | Command-line tool to interact with Docker     |
| Docker Daemon    | Background process managing Docker containers |
| Docker Hub       | Online repository to store and share images   |

#### **Demonstration of Running a Container using Docker (Theoretical Steps)**

1. Install Docker
  - i. Download and install Docker Desktop from <https://www.docker.com>.
  - ii. Start Docker on your local machine.
  - iii. Verify installation by running:

*docker --version*

2. Pull a Base Image
  - i. Use Docker Hub to pull a popular image (e.g., Ubuntu, Nginx, Python):

*docker pull ubuntu*

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### 3. Run a Container from an Image

- i. Use the docker run command:

```
docker run -it ubuntu
```

- ii. This opens an interactive terminal session in the container.
- iii. Run Linux commands inside the container (e.g., ls, pwd, apt update).

### 4. Exit the Container

- i. Type exit to close the session and stop the container.

### 5. List Running and Stopped Containers

- i. View active containers:

```
docker ps
```

- ii. View all containers (including stopped ones):

```
docker ps -a
```

### 6. Remove Containers

- i. Stop the container (if still running):

```
docker stop <container_id>
```

- ii. Remove the container:

```
docker rm <container_id>
```

### 7. Run a Web Server Container (Optional)

- i. Run a web server (e.g., Nginx):

```
docker run -d -p 8080:80 nginx
```

- ii. Open <http://localhost:8080> in browser to see the Nginx welcome page.

### Use Case Example:

- ❖ Running a Linux environment on any OS for testing
- ❖ Hosting a web server like Apache or Nginx inside a container
- ❖ Developing and testing Python, Node.js, or Java apps
- ❖ Containerizing databases like MySQL or MongoDB for quick use
- ❖ Experimenting with new tools and languages without affecting host OS
- ❖ Education and training in DevOps or system admin practices
- ❖ Deploying a standalone app for development or demonstration

### Implementation:

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```
ubuntu@ip-172-31-40-218: ~  
ubuntu@ip-172-31-40-218:~$  
ubuntu@ip-172-31-40-218:~$ systemctl status docker  
● docker.service - Docker Application Container Engine  
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)  
   Active: active (running) since Mon 2023-04-10 19:46:05 UTC; 18min ago  
 TriggeredBy: ● docker.socket  
     Docs: https://docs.docker.com  
    Main PID: 684 (dockerd)  
      Tasks: 12  
     Memory: 141.2M  
        CPU: 3.736s  
    CGroup: /system.slice/docker.service  
            └─684 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock  
  
Apr 10 19:46:01 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:01.899794323Z" level=info msg="ccResolverWrapper: se  
Apr 10 19:46:01 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:01.899956895Z" level=info msg="ClientConn switching  
Apr 10 19:46:02 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:02.339992511Z" level=info msg="[graphdriver] using p  
Apr 10 19:46:03 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:03.665306795Z" level=info msg="Loading containers: s  
Apr 10 19:46:04 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:04.873139021Z" level=info msg="Default bridge (docke  
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.081644328Z" level=info msg="Loading containers: d  
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.543882435Z" level=info msg="Docker daemon" commit  
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.547797680Z" level=info msg="Daemon has completed  
Apr 10 19:46:05 ip-172-31-40-218 systemd[1]: Started Docker Application Container Engine.  
Apr 10 19:46:05 ip-172-31-40-218 dockerd[684]: time="2023-04-10T19:46:05.743749833Z" level=info msg="API listen on /run/do  
lines 1-22/22 (END)
```

```
ubuntu@ip-172-31-40-218:~$ docker ps  
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES  
ubuntu@ip-172-31-40-218:~$ docker images  
REPOSITORY    TAG       IMAGE ID   CREATED   SIZE  
ubuntu@ip-172-31-40-218:~$
```

```
ubuntu@ip-172-31-40-218:~$  
ubuntu@ip-172-31-40-218:~$ pwd  
/home/ubuntu  
ubuntu@ip-172-31-40-218:~$ mkdir my-website  
ubuntu@ip-172-31-40-218:~$ cd my-website/  
ubuntu@ip-172-31-40-218:~/my-website$ wget https://www.free-css.com/assets/files/free-css-templates/download/page290/wave-cafe.zip  
--2023-04-10 20:06:14-- https://www.free-css.com/assets/files/free-css-templates/download/page290/wave-cafe.zip  
Resolving www.free-css.com (www.free-css.com)... 217.160.0.242, 2001:8d8:100f:f000::28f  
Connecting to www.free-css.com (www.free-css.com)|217.160.0.242|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 11896390 (11M) [application/zip]  
Saving to: 'wave-cafe.zip'  
  
wave-cafe.zip      100%[=====>] 11.34M  6.08MB/s  in 1.9s  
  
2023-04-10 20:06:17 (6.08 MB/s) - 'wave-cafe.zip' saved [11896390/11896390]  
  
ubuntu@ip-172-31-40-218:~/my-website$  
ubuntu@ip-172-31-40-218:~/my-website$  
ubuntu@ip-172-31-40-218:~/my-website$ ls  
wave-cafe.zip  
ubuntu@ip-172-31-40-218:~/my-website$ unzip wave-cafe.zip
```

```
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.ttf  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.woff  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-regular-400.woff2  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.eot  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.svg  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.ttf  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.woff  
inflating: 2121_wave_cafe/fontawesome/webfonts/fa-solid-900.woff2  
creating: 2121_wave_cafe/img/  
inflating: 2121_wave_cafe/img/about-1.png  
inflating: 2121_wave_cafe/img/about-2.png  
inflating: 2121_wave_cafe/img/hot-americano.png  
inflating: 2121_wave_cafe/img/hot-cappuccino.png  
inflating: 2121_wave_cafe/img/hot-espresso.png  
inflating: 2121_wave_cafe/img/hot-latte.png  
inflating: 2121_wave_cafe/img/iced-americano.png  
inflating: 2121_wave_cafe/img/iced-cappuccino.png  
inflating: 2121_wave_cafe/img/iced-espresso.png  
inflating: 2121_wave_cafe/img/iced-latte.png  
inflating: 2121_wave_cafe/img/smoothie-1.png  
inflating: 2121_wave_cafe/img/smoothie-2.png  
inflating: 2121_wave_cafe/img/smoothie-3.png  
inflating: 2121_wave_cafe/img/smoothie-4.png  
inflating: 2121_wave_cafe/img/special-01.jpg  
inflating: 2121_wave_cafe/img/special-02.jpg  
inflating: 2121_wave_cafe/img/special-03.jpg  
inflating: 2121_wave_cafe/img/special-04.jpg  
inflating: 2121_wave_cafe/img/special-05.jpg  
inflating: 2121_wave_cafe/img/special-06.jpg  
inflating: 2121_wave_cafe/index.html  
creating: 2121_wave_cafe/js/
```

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```
ubuntu@ip-172-31-40-218: ~/my-website
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe  wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ cd 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ ls
css  fontawesome  img  index.html  js  video
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cp -R * ../
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cd ..
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe  css  fontawesome  img  index.html  js  video  wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ rm -rf wave-cafe.zip 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
css  fontawesome  img  index.html  js  video
ubuntu@ip-172-31-40-218:~/my-website$ nano Dockerfile
```

```
GNU nano 6.2 Dockerfile
FROM httpd:2.4
COPY . /usr/local/apache2/htdocs/

[ Wrote 2 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^V Paste      ^J Justify    ^/ Go To Line  M-E Redo      M-C Copy
```



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```
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ cd 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ ls
css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cp -R * ../
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$
ubuntu@ip-172-31-40-218:~/my-website/2121_wave_cafe$ cd ..
ubuntu@ip-172-31-40-218:~/my-website$ ls
2121_wave_cafe css fontawesome img index.html js video wave-cafe.zip
ubuntu@ip-172-31-40-218:~/my-website$ rm -rf wave-cafe.zip 2121_wave_cafe
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ ls
css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website$ nano Dockerfile
ubuntu@ip-172-31-40-218:~/my-website$ ls
Dockerfile css fontawesome img index.html js video
ubuntu@ip-172-31-40-218:~/my-website$ docker build . -t my-website:latest
Sending build context to Docker daemon 13.61MB
Step 1/2 : FROM httpd:2.4
2.4: Pulling from library/httpd
f1f26f570256: Pull complete
a6b093ae1967: Pull complete
6b400bbb27df: Pull complete
6e310dd059b6: Pull complete
471cb5914961: Pull complete
Digest: sha256:4055b18d92fd006f74d4a2aac172a371dc9a750eaa78000756dee55a9beb4625
Status: Downloaded newer image for httpd:2.4
--> dcl95e13784
Step 2/2 : COPY . /usr/local/apache2/htdocs/
--> 7d48427f5e2f
Successfully built 7d48427f5e2f
Successfully tagged my-website:latest
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ clear
```

```
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
my-website latest 7d48427f5e2f 15 seconds ago 159MB
httpd 2.4 dcl95e13784 4 days ago 145MB
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker run -d -p 80:80 my-website:latest
e0a6d7f3ab6718alb648d9b5f00dcc89e846d1fe12bd568ce9b1412fc0d3c9da
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$
ubuntu@ip-172-31-40-218:~/my-website$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
e0a6d7f3ab67 my-website:latest "httpd-foreground" 8 seconds ago Up 7 seconds 0.0.0.0:80->80/tcp, :::80->80/tcp
trusting_rosalind
ubuntu@ip-172-31-40-218:~/my-website$
```

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Best selling ice cream brand where we sell ice cream made from fresh fruits and milk.



**Conclusion:** We have successfully understood Dockerfile instructions, build an image for a sample web application using DOCKERFILE.

**LO Mapping:** *LO is mapped*